

Assignment 6 (Bonus assignment) – Android APK Build & Deployment

Objective:

Transition of the Assignment 5 Unity project from the editor to a functional Android mobile application.

Brief Implementation Note:

The Assignment 5 Unity and Vuforia AR project was converted into an Android mobile application using Unity Build Profiles. The platform was switched to Android and the Player Settings were configured using IL2CPP scripting backend, ARM64 architecture, and Android API Level 29 for compatibility with modern Android devices.

The Vuforia AR Camera and Image Target settings were optimized for stable marker tracking and augmentation. OpenGL ES3 was used as the primary Graphics API for improved mobile rendering performance. When the application is opened, camera permission must be allowed to enable marker detection. After granting permission, the custom marker image attached with the project can be scanned using the mobile camera, and the rotating 3D Earth globe appears augmented on top of the marker.

The APK was generated using Unity Build Profiles and installed manually on an Android smartphone through sideload installation. The application was tested successfully on an Android 15 device, where the Earth globe was tracked and augmented correctly with stable real-time rotation.

Steps Followed to Export APK:

1. Opened the Unity AR project.
2. Switched the platform to Android using Build Profiles.
3. Configured Android Player Settings and Vuforia settings.
4. Set IL2CPP as the scripting backend and enabled ARM64 architecture.
5. Generated the APK using the Build option.
6. Installed the APK on the Android smartphone.
7. Allowed camera permission and scanned the custom marker image to view the rotating Earth augmentation.